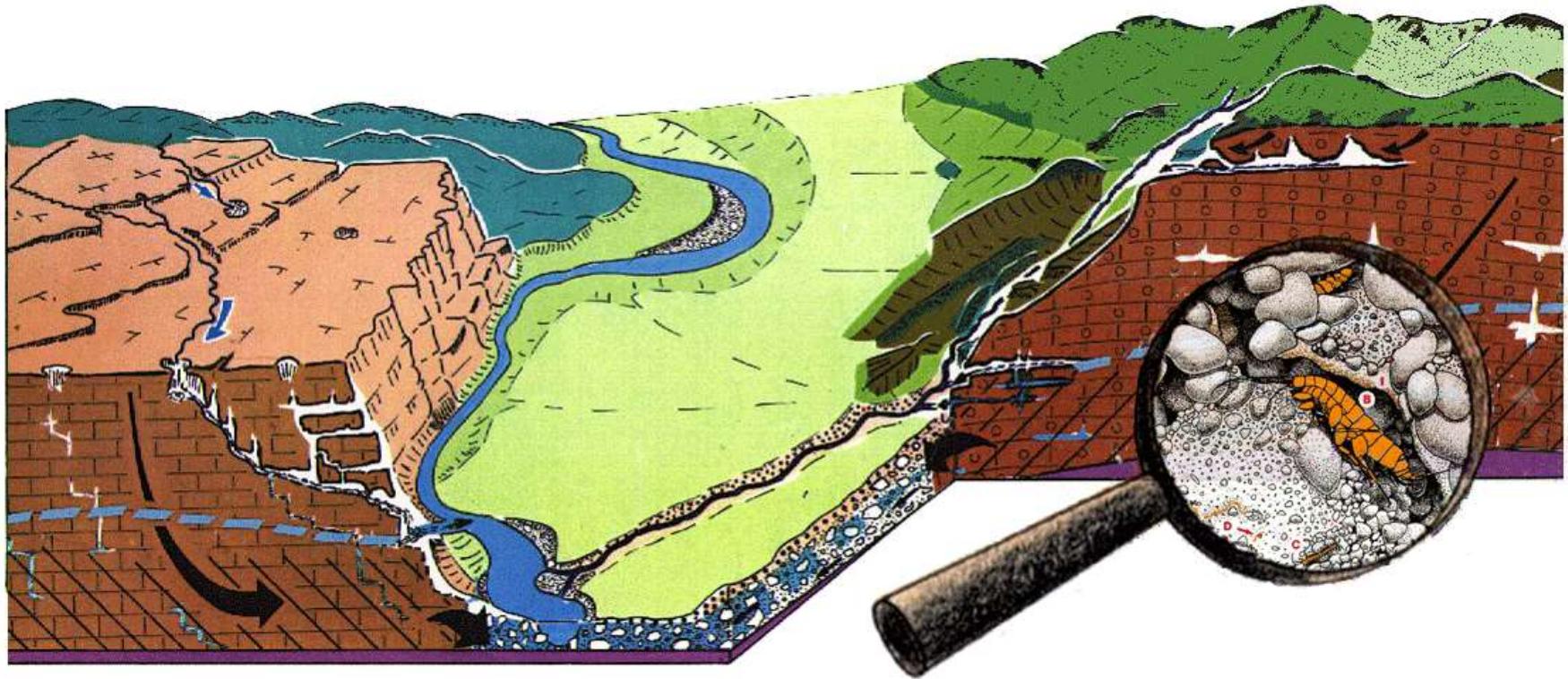


Characterisation of groundwater habitats: the faunal perspective



Florian MALARD

CNRS / University of Lyon, France

Scope

→ Bringing groundwater ecosystems into the WFD

(20) **Research** should be conducted in order to provide better **criteria** for ensuring groundwater ecosystem quality and protection. Where necessary, the findings obtained should be taken into account when implementing or revising this Directive.

DIRECTIVE 2006/118/EC

→ Ten years later, do we know more ?

→ Can we provide habitat typology and habitat-specific biological references ?

Content

Definitions

Habitat typology

History of groundwater habitat typology

Habitat typology: EU / National

Scoring habitat quality

Habitat-specific biological references

History of species occurrence data sets

Species richness

Community composition

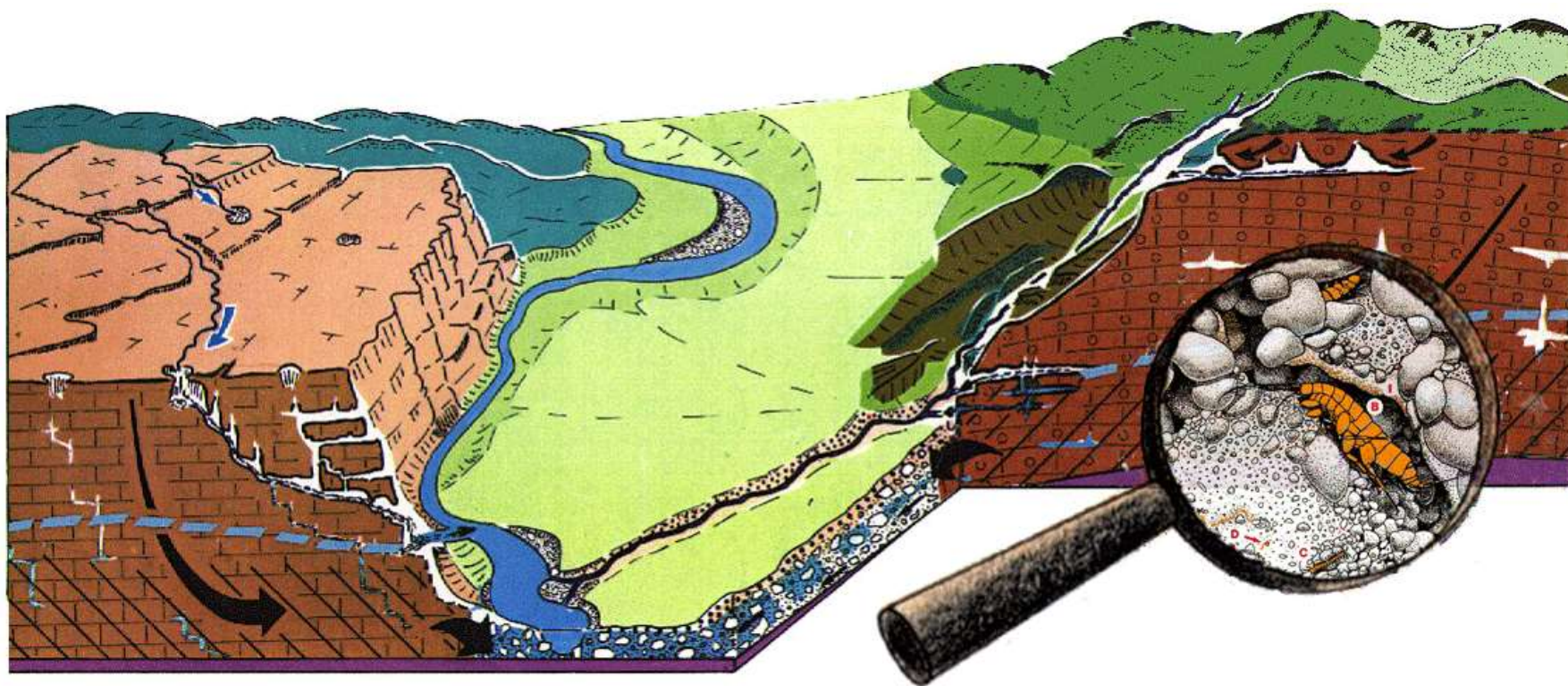
Take home messages



Definitions

Habitat	The environment in which an organism / a community of organisms lives	
Habitat type:	A characterization of a collection of spatial entities that are considered equivalent as habitats.	Criteria & map
Habitat potential:	The potential for an habitat to harbor an organism / a community of organisms	Scoring habitats
Reference condition:	The value of a biological quality element at high ecological status	Species data base
Ecological status:	The deviation of a biological quality element from the reference condition. e.g. “The level of diversity of invertebrate taxa shows no sign of alteration from undisturbed levels” (WFD)	

Habitat typology



History of groundwater habitat typology

Until the 60s'

Cave
habitats



From the 90s'

Fractured

Porous

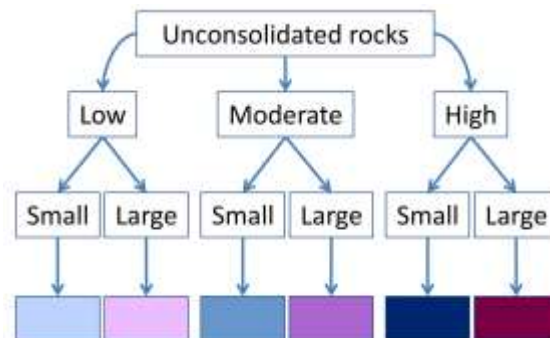
Karst



Since 21st century

Permeability

Pore size



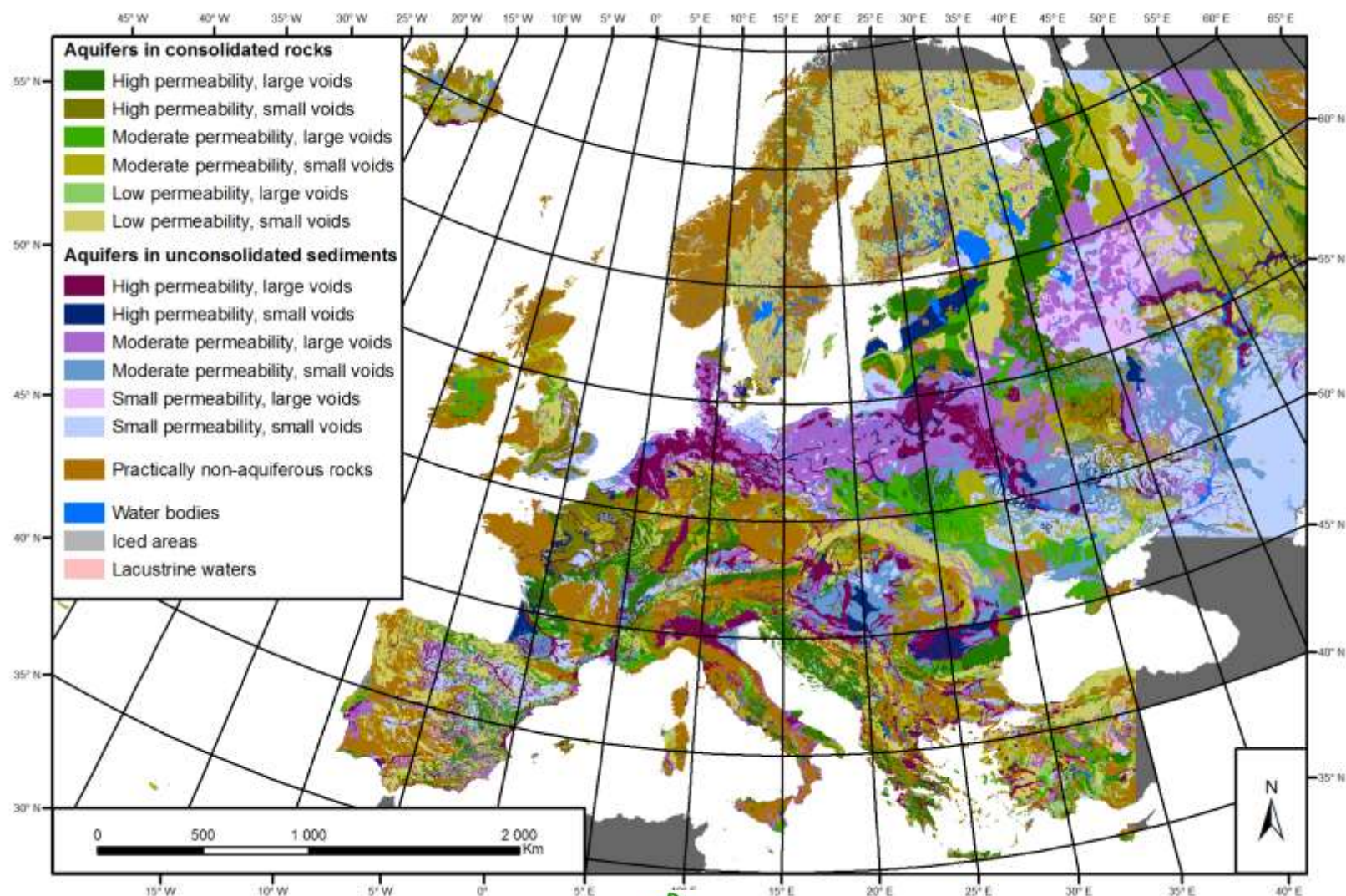
A consensus for eco-hydrogeological criteria for habitat typology

Nine convergent proposals since 2009

Hahn (2009) *Hydrogeology Journal*; Tomlinson & Boulton (2010). *Marine and Freshwater Research*; Larned (2012) *Freshwater Biology*; Cornu et al. (2013) *Hydrogeology Journal*; Gutjahr et al. (2014) *Subterranean Biology*; Weitowitz et al. (2017) *Hydrogeology Journal*

Pore space, permeability, recharge mode

Habitat typology at EU level



Editor's choice

Cornu *et al.* 2013

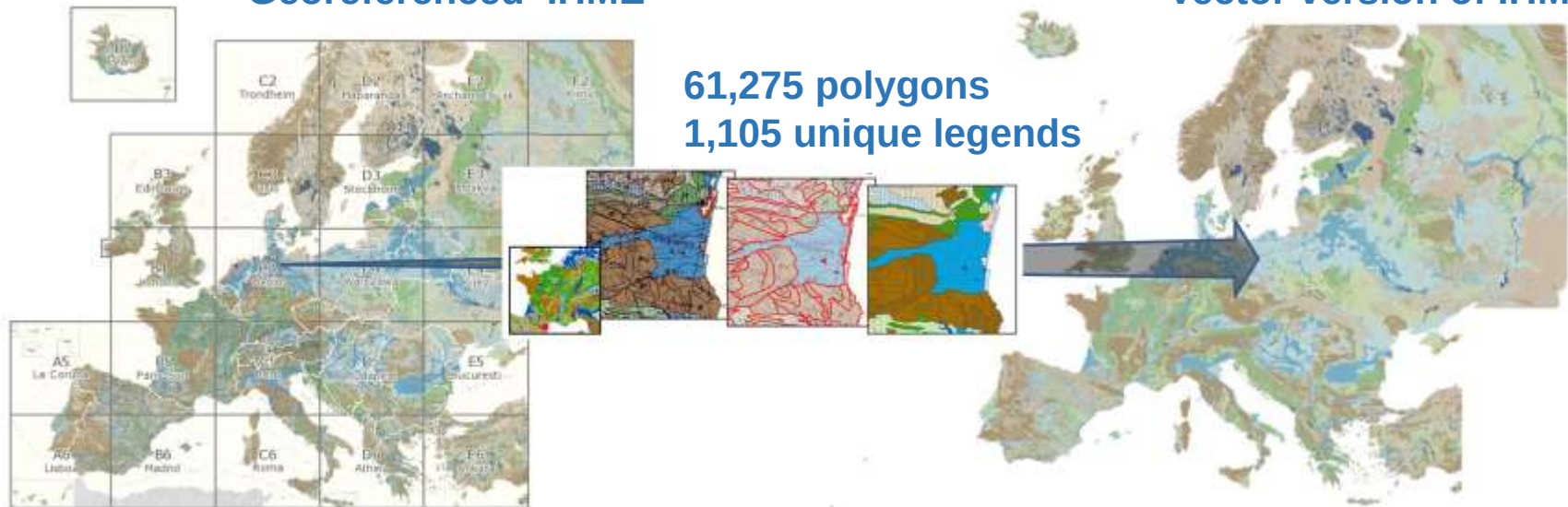


Global Freshwater Biodiversity Atlas

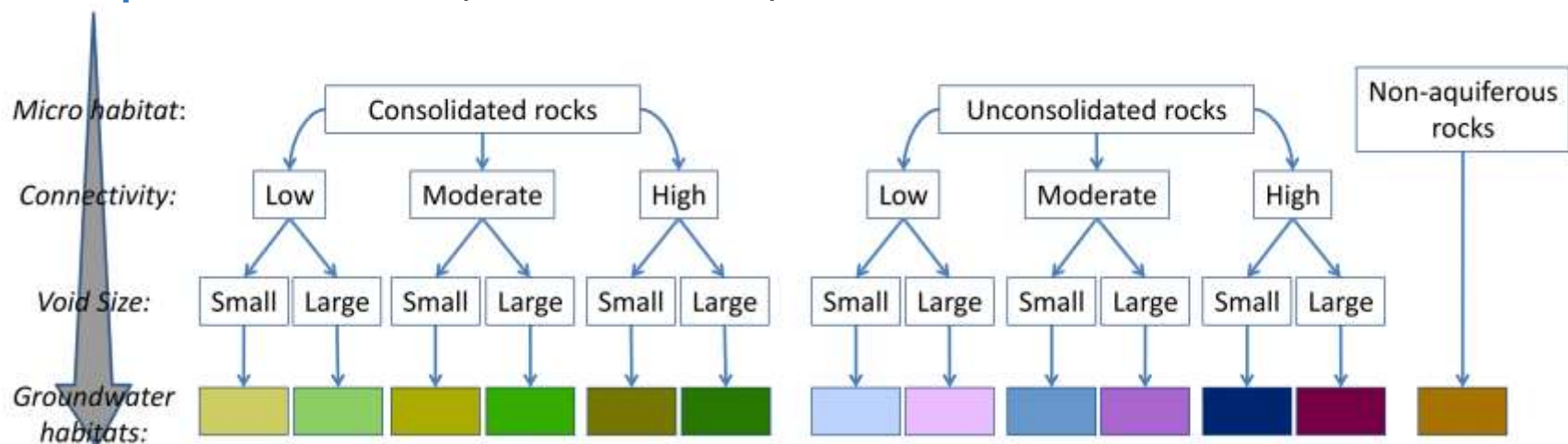
The groundwater habitat map of Europe (GHME)

Georeferenced IHME

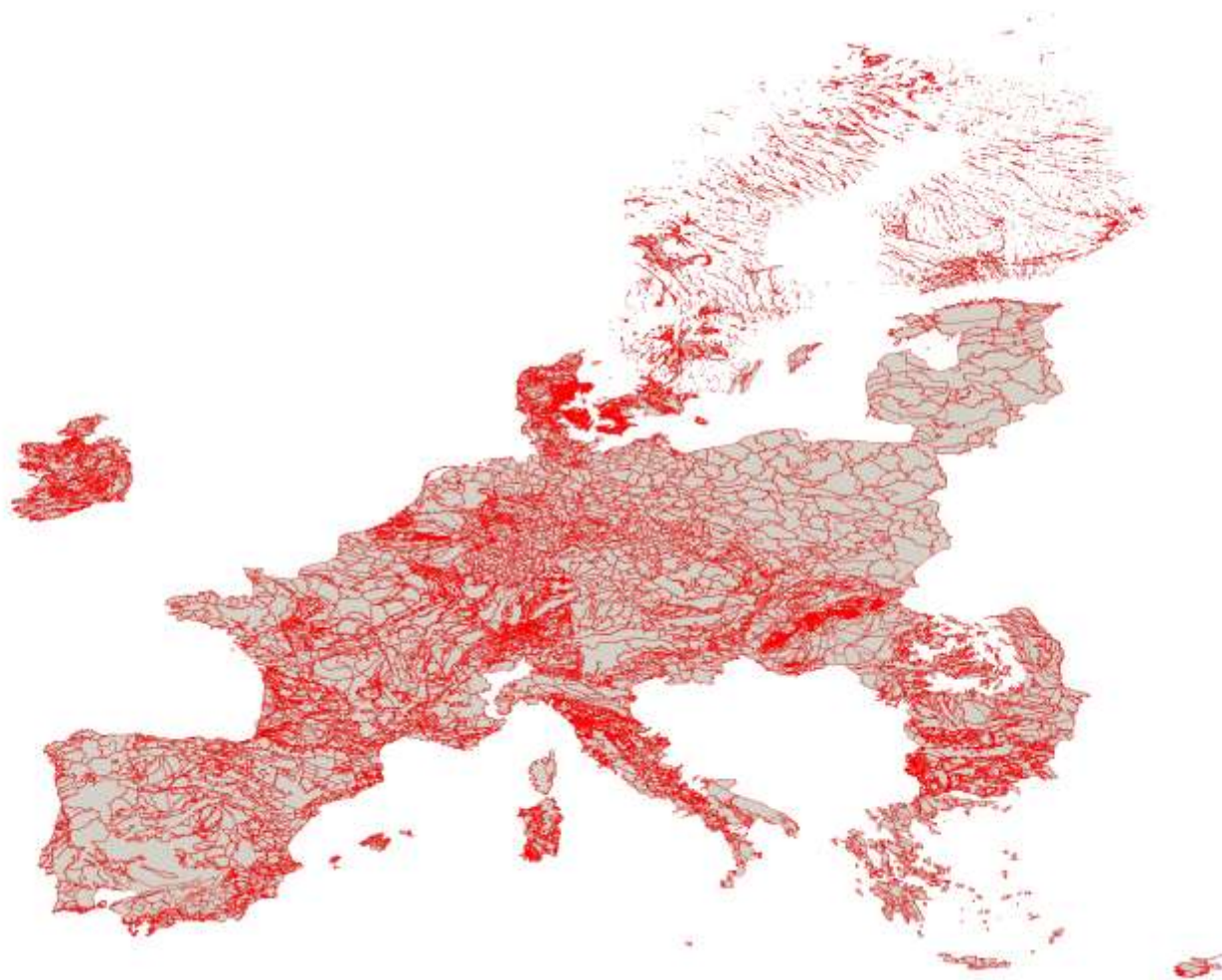
Vector version of IHME



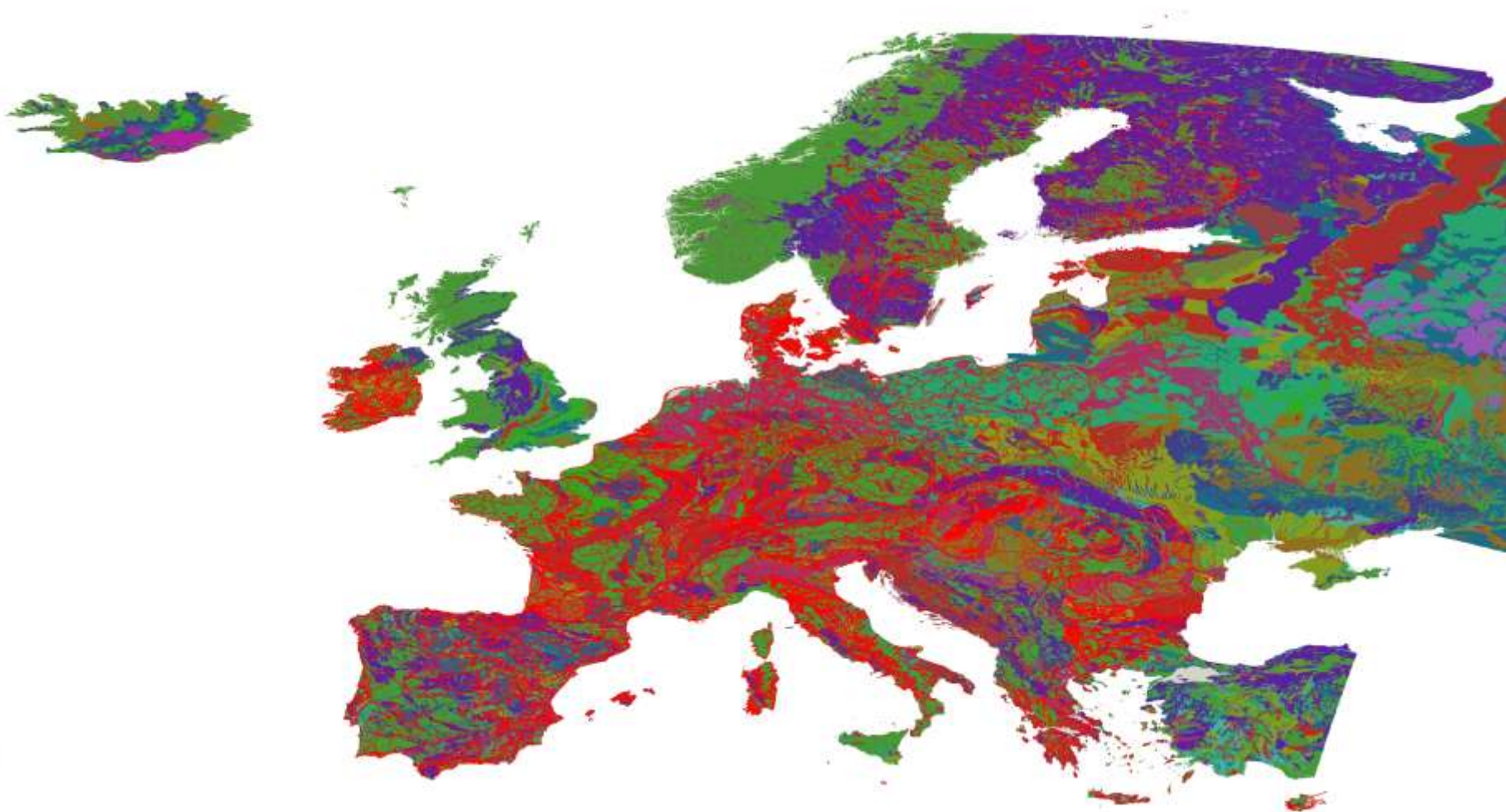
From aquifers to habitats (Cornu et al. 2013)



EU Map of groundwater bodies



EU Map of groundwater bodies with habitat attributes



Refining habitat typology at national level (UK)



Weitowitz *et al.* (2017, in revision)

Twelve geo-habitats

Among karstic, porous, mixed-sandstone and fractured aquifers

Criteria and data collection

Transmissivity (m^2/d): 1725 pumping test values

Porosity (%): 519 core sample values

DO, DOC, nitrate, calcium (thousands values)

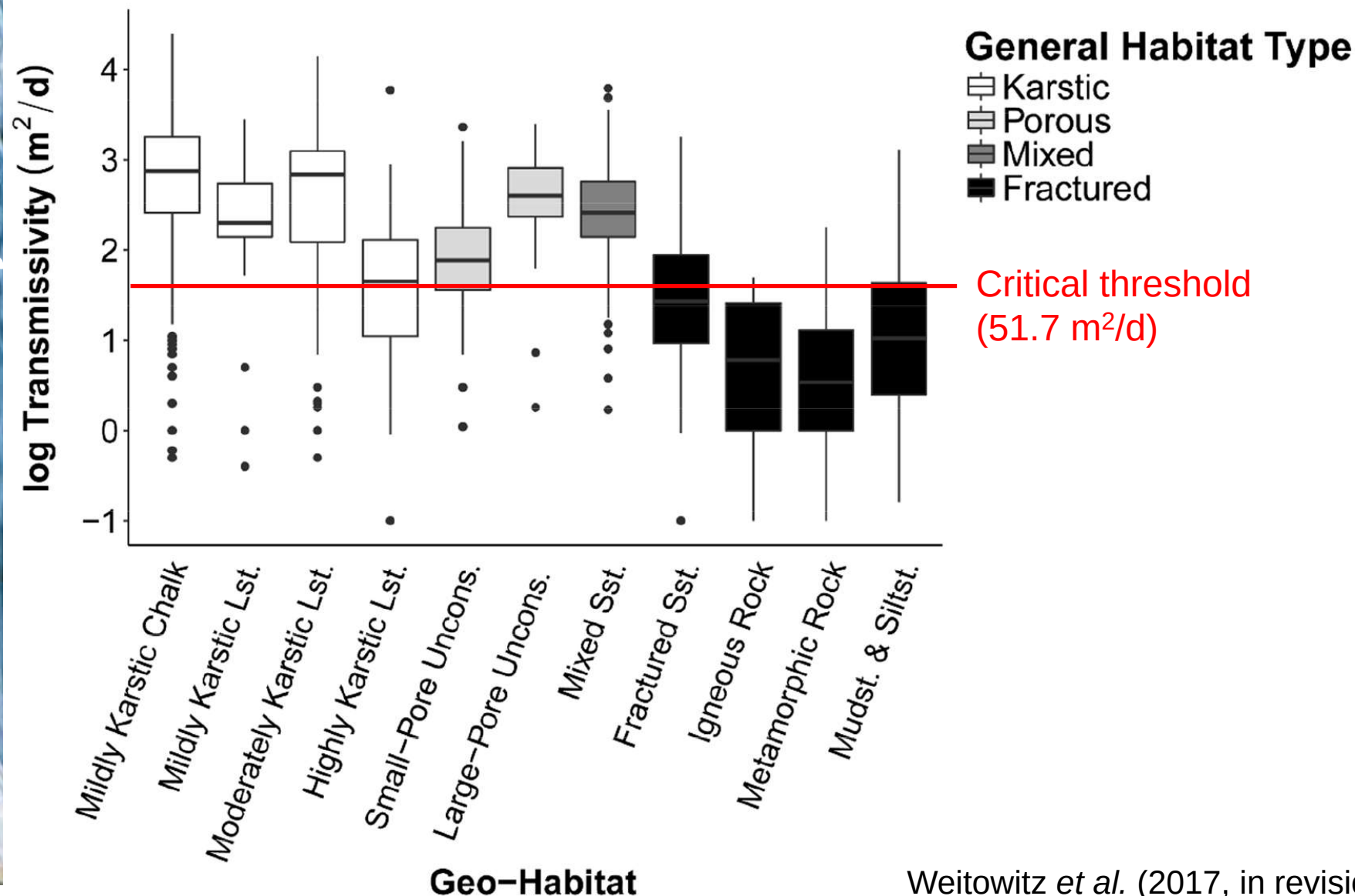
Cave development, habitat space

Geo-habitat quality

Habitat score using critical threshold for each criteria

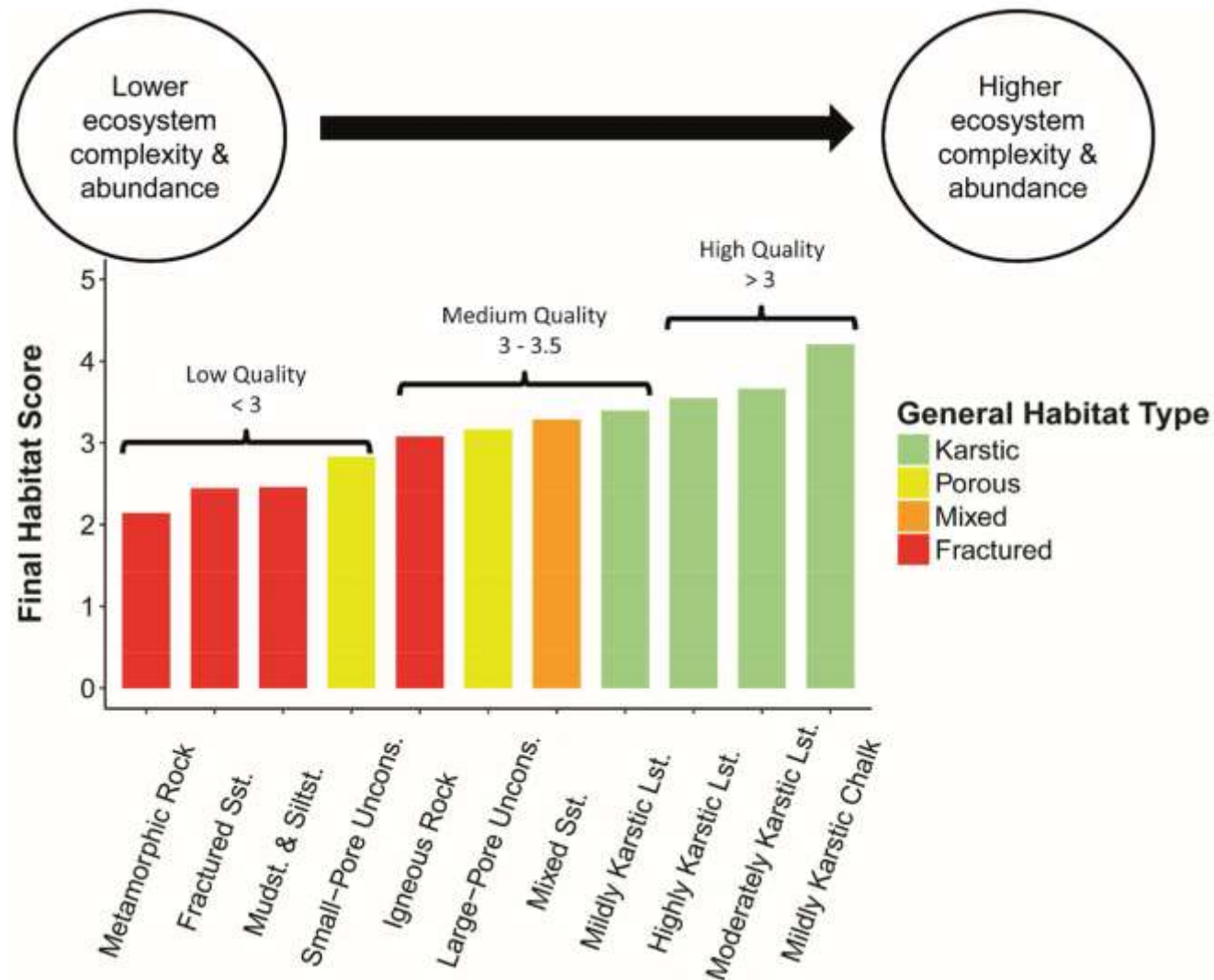


Critical threshold for transmissivity

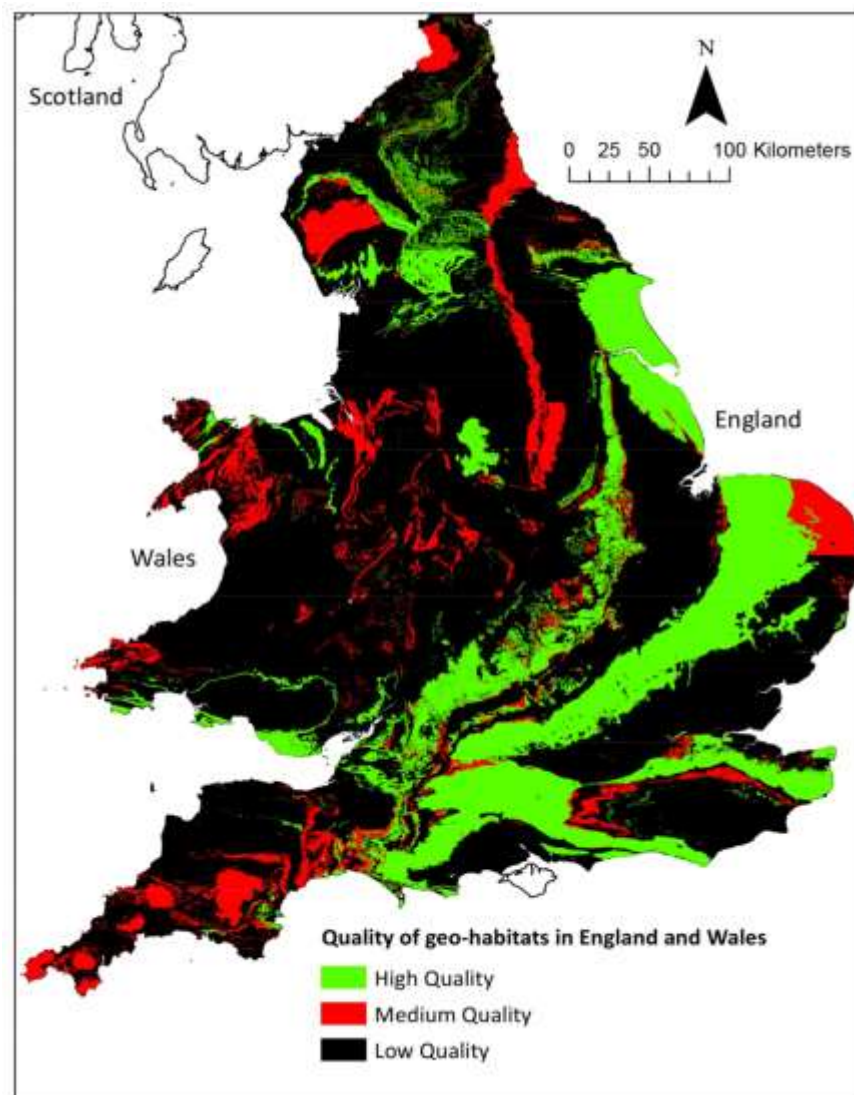


Weitowitz *et al.* (2017, in revision)

Scoring habitat quality

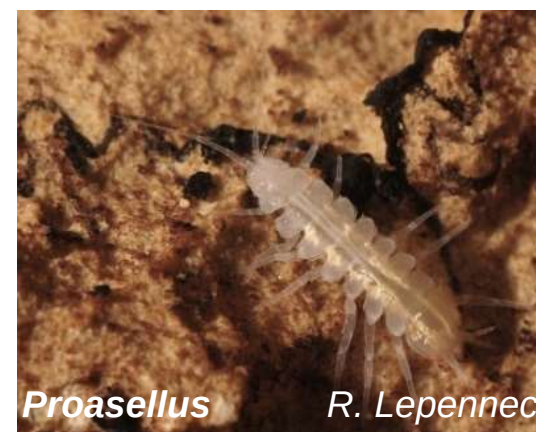


Mapping habitat quality



Weitowitz *et al.* (2017, in revision)

Habitat-specific biological references



History of species occurrence data sets

Prior to 2000

➡ National Initiatives



*Mapping
Subterranean
Biodiversity*

2002-2014

➡ European Projects



PASCALIS

EU FP5 2002-2004



EU FP7 2009-2014

2015...

➡ DNA-based methods



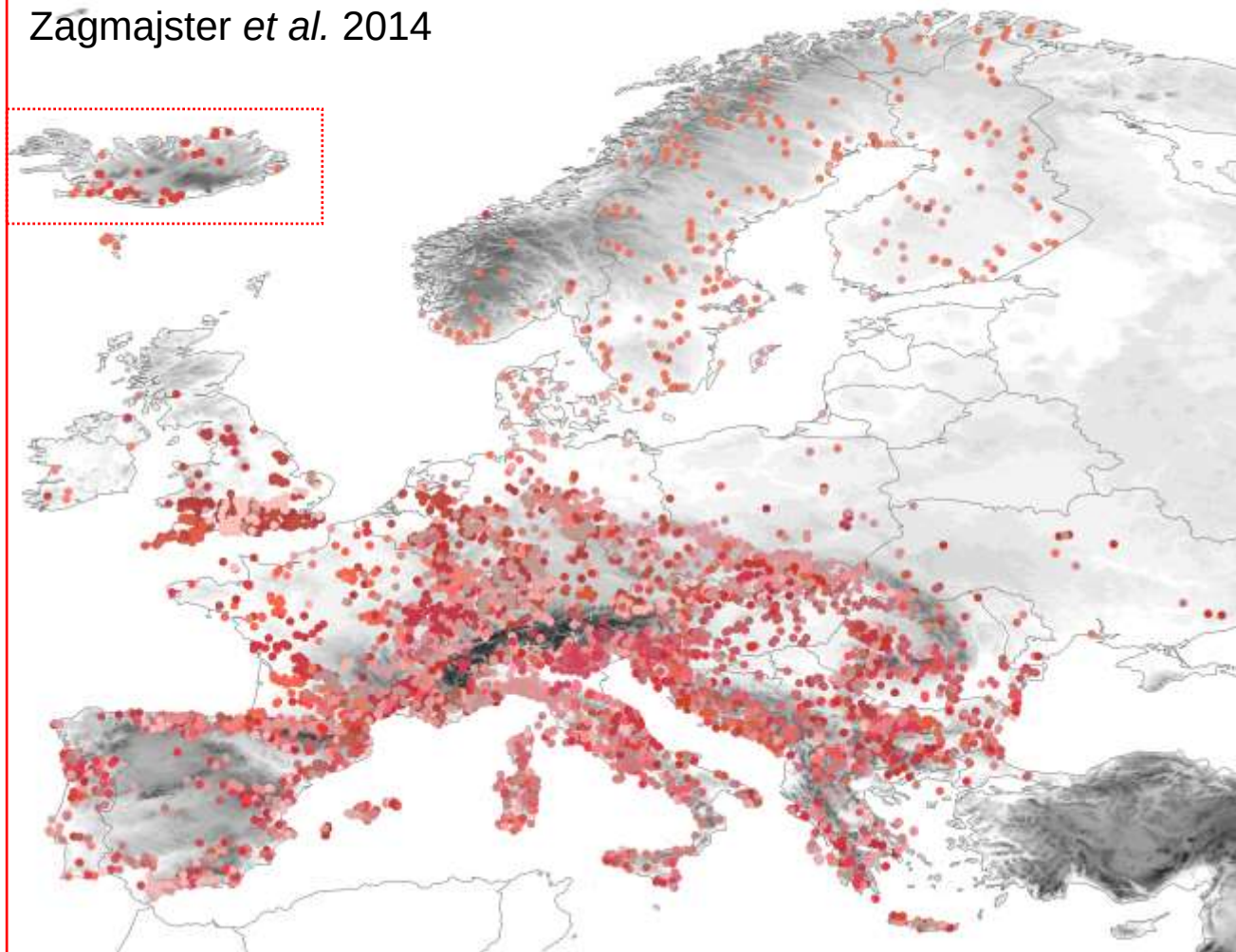
PROTEUS



The European groundwater crustacean data set (EGCD)

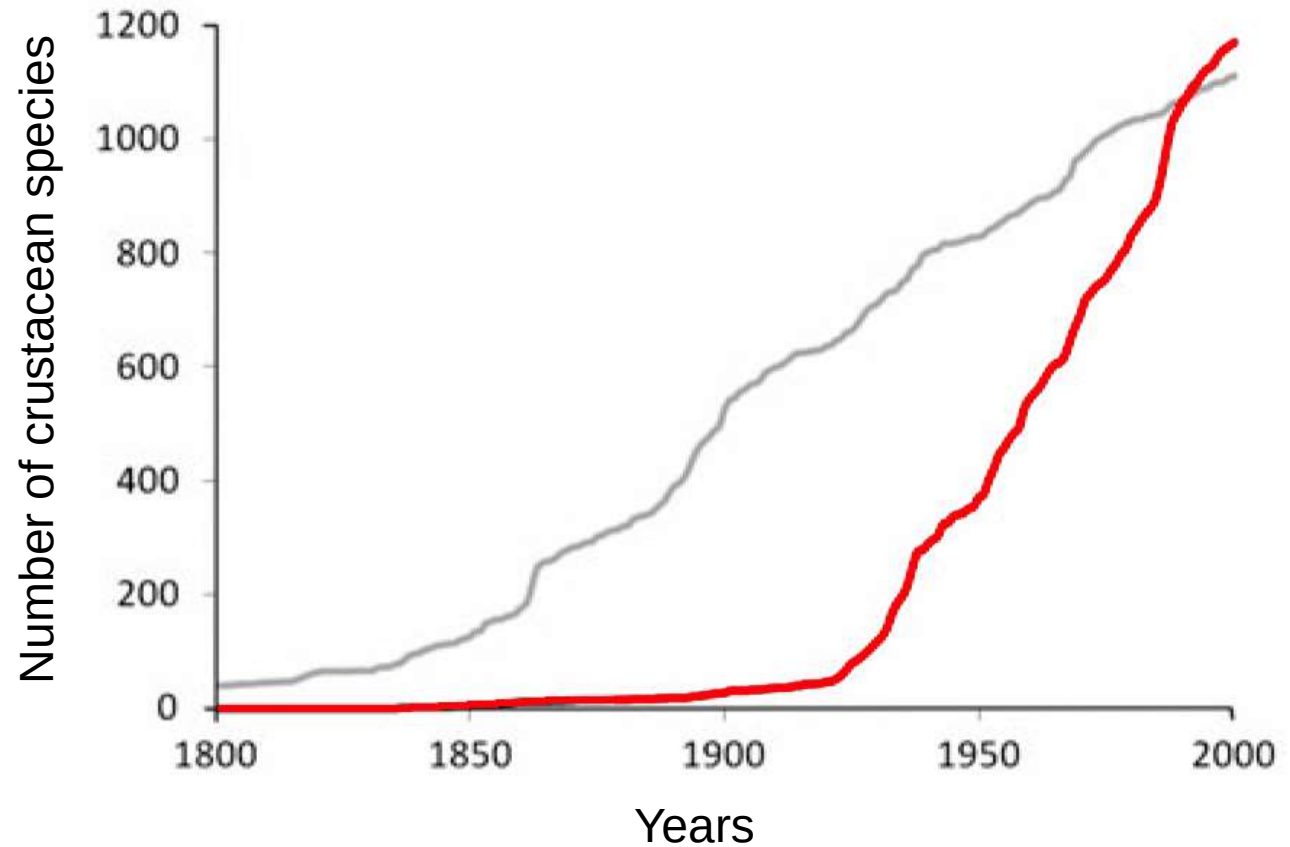


Zagmajster et al. 2014



21,700 records, 12 orders, 46 families, 165 genera and 1570 species of groundwater Crustacea

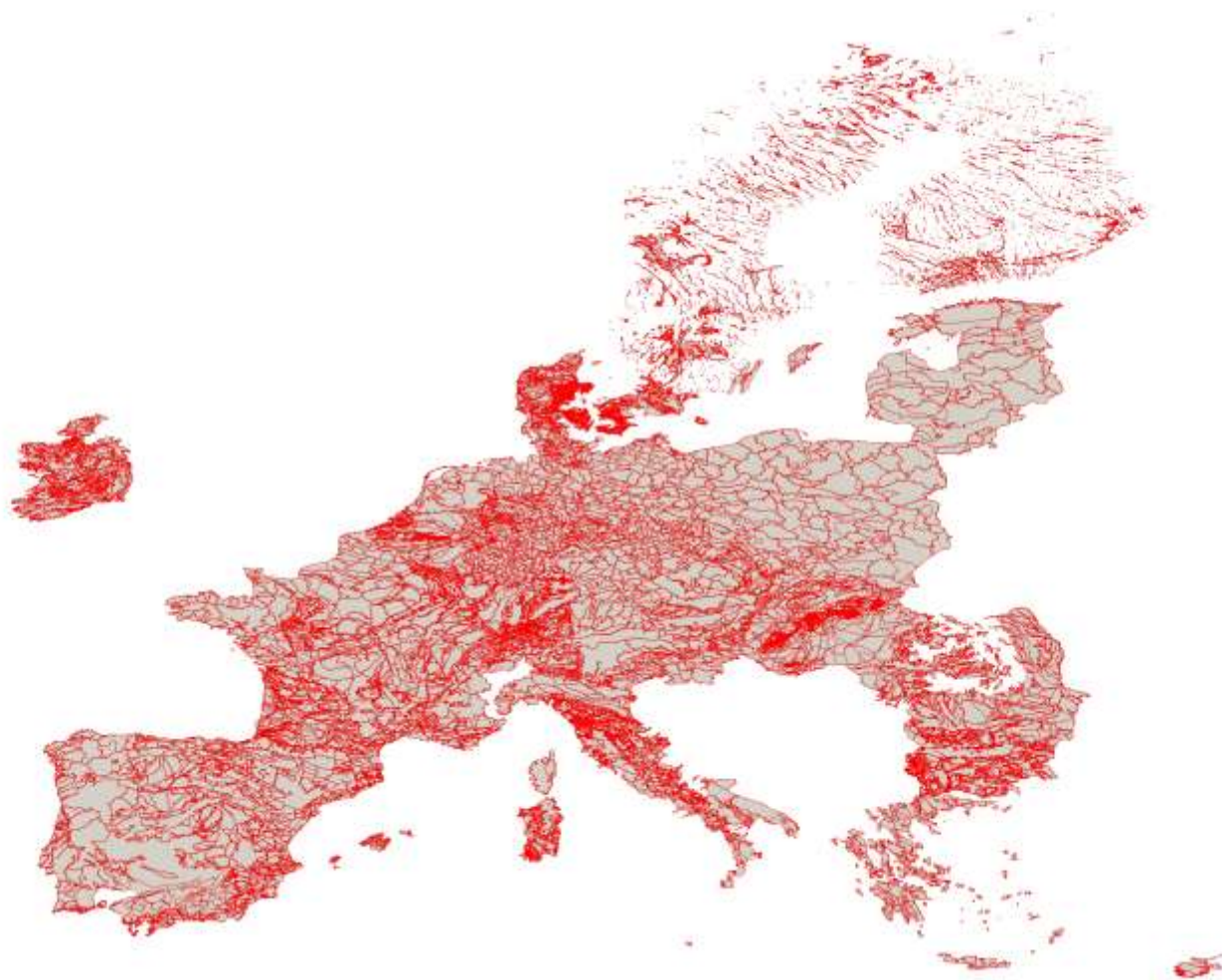
Groundwater contains more crustaceans than surface water



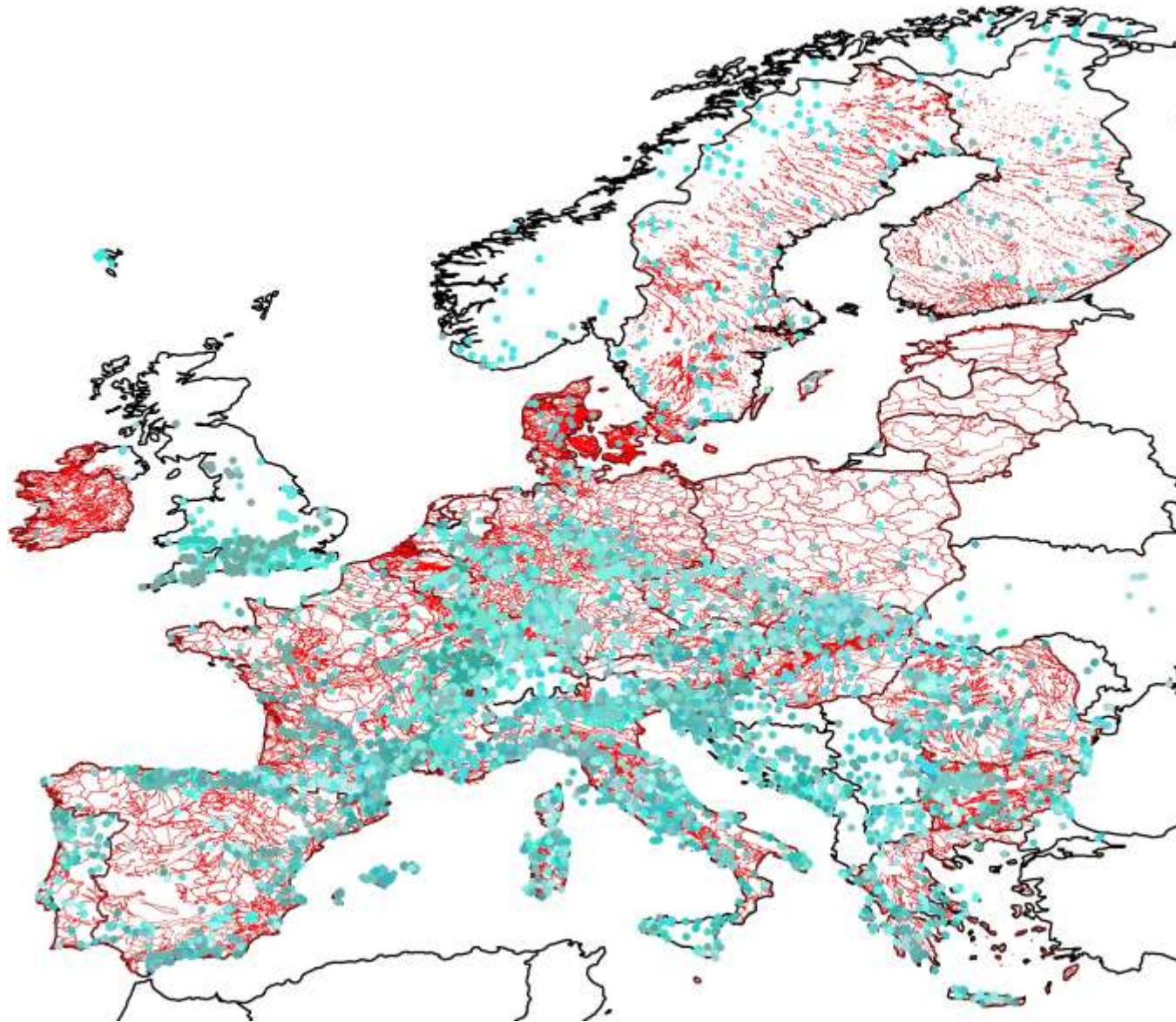
Hydrobiologia
Springer

Stoch & Galassi 2010

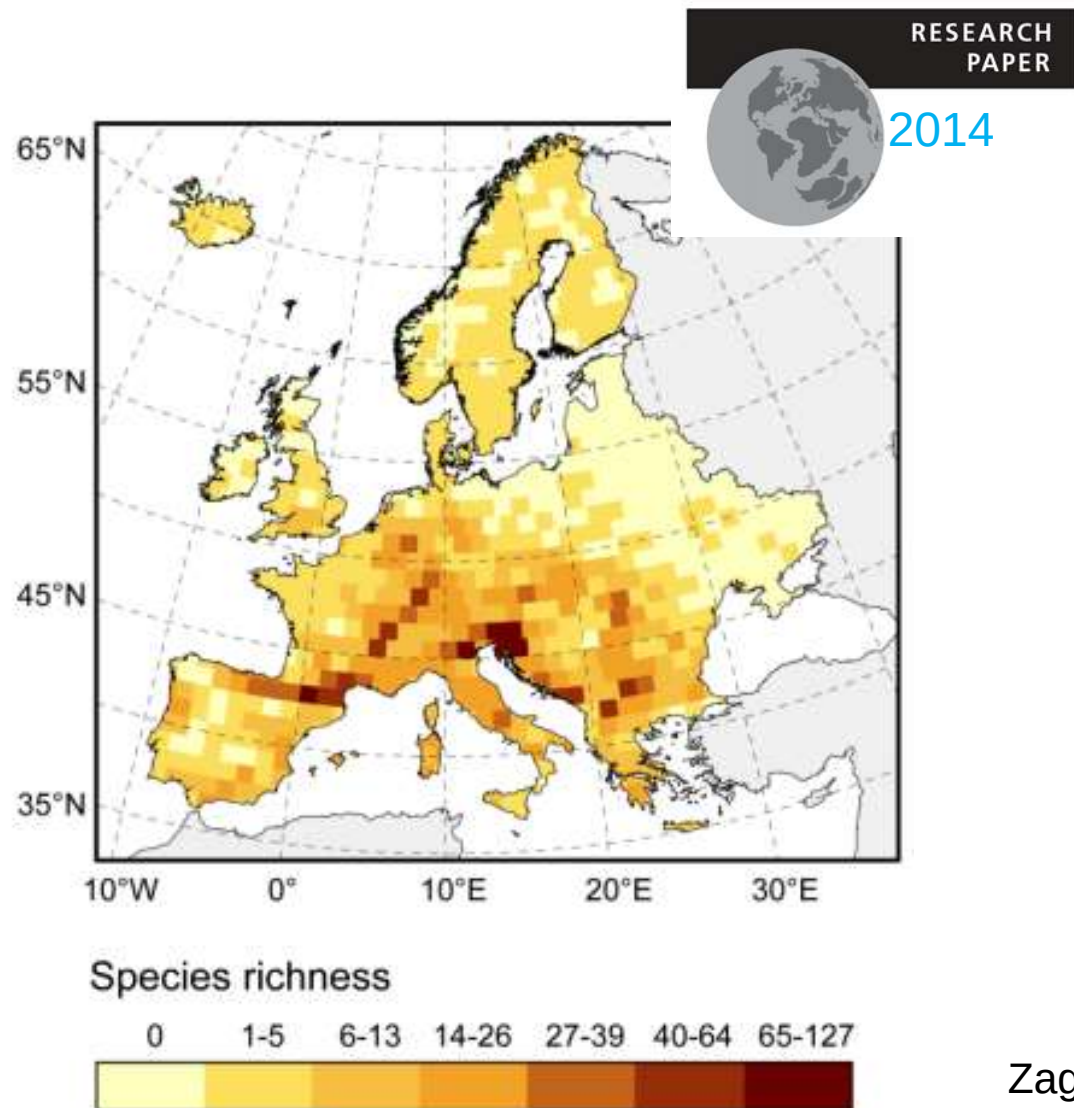
EU Map of groundwater bodies



EU Map of groundwater bodies with species content



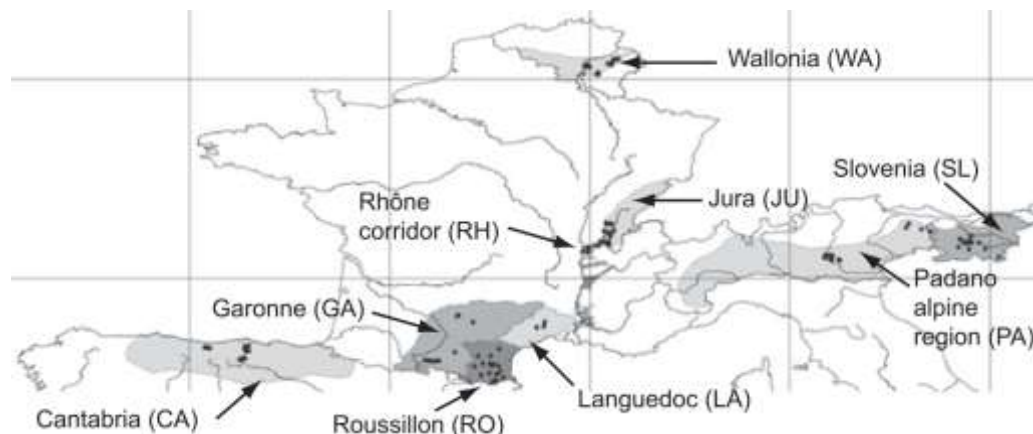
Variation in species richness among regions



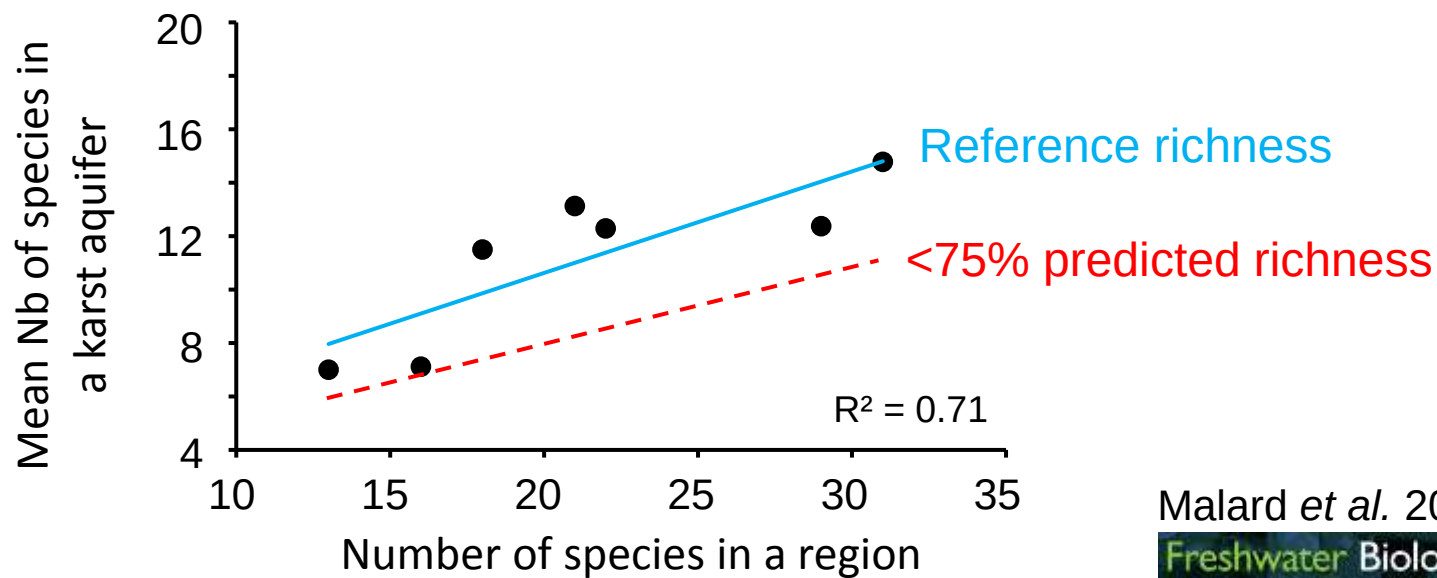
Zagmajster *et al.* 2014

Species richness as a biological reference

9 regions in Europe, several « high ecological status » karst aquifers



Setting the « karst habitat » reference species richness

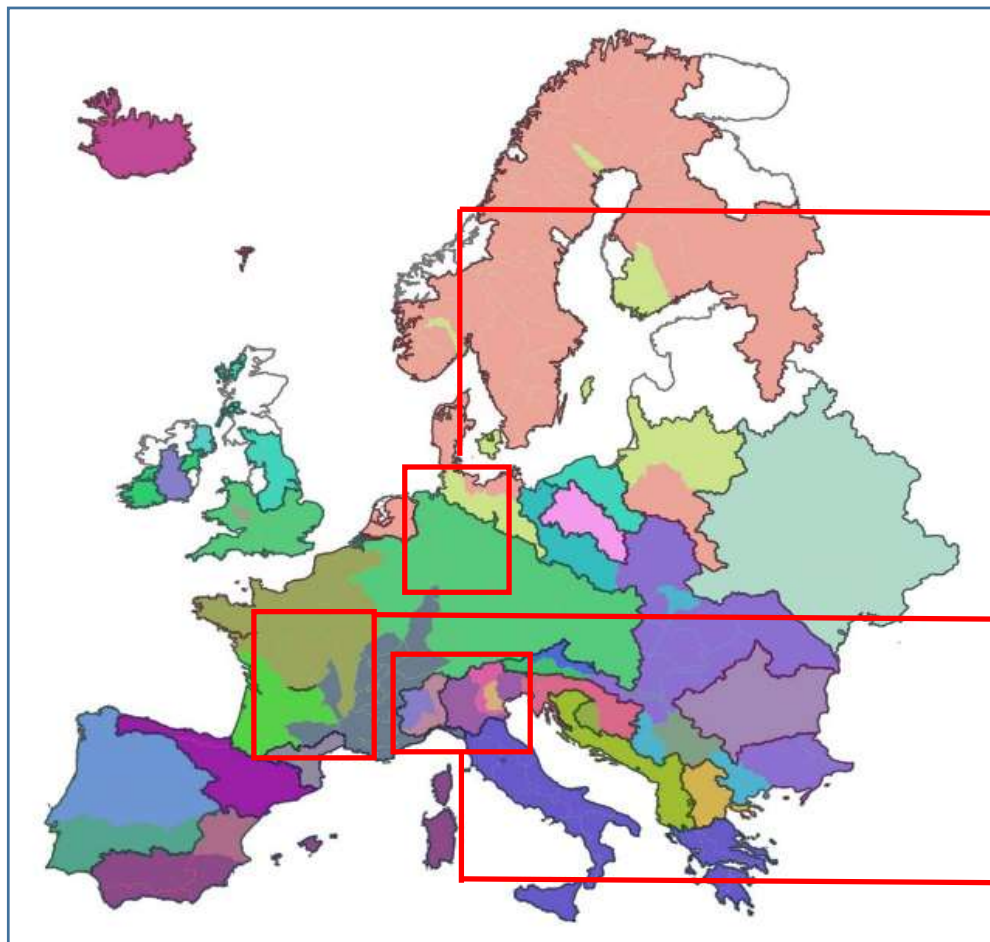


Malard *et al.* 2009
Freshwater Biology

Community composition: using groundwater bioregions

Groundwater bioregions of Europe

Regional refinements



Stein *et al.* (2012)



Symposium Book

Ferreira *et al.* (2005)

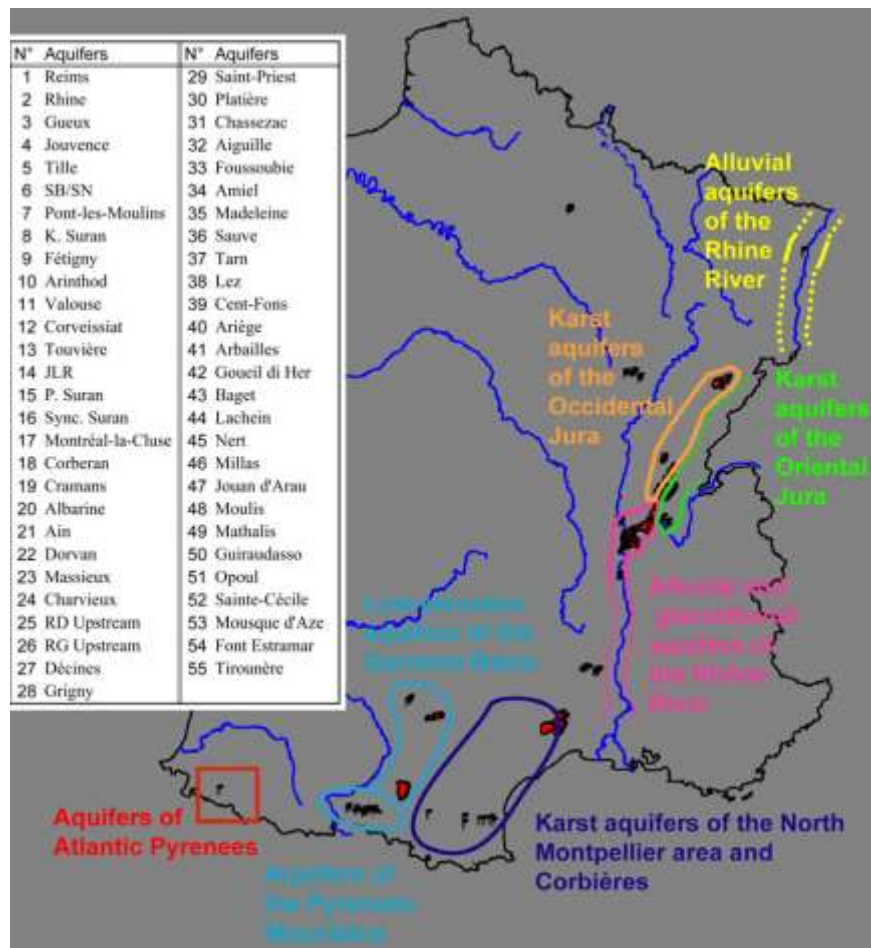


Stoch & Galassi 2010

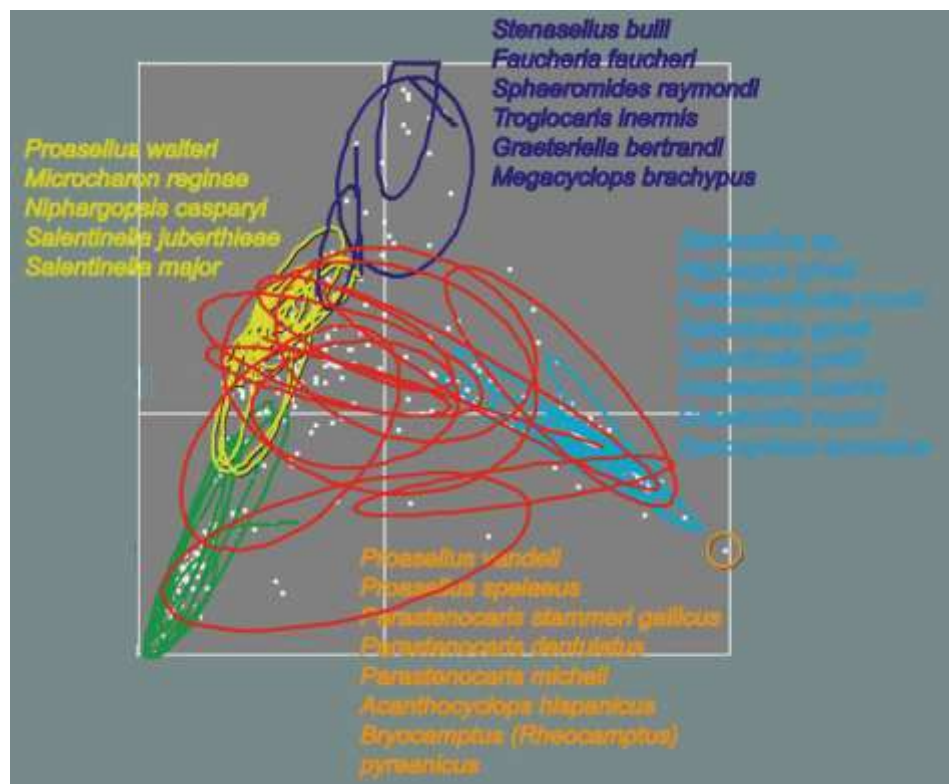
Fuzzy statistics applied to EGCD*
Stoch *et al.* (in progress)

Community composition: using bioregions (France)

55 aquifers, 380 species → 8 bioregions



Sets of characteristic species

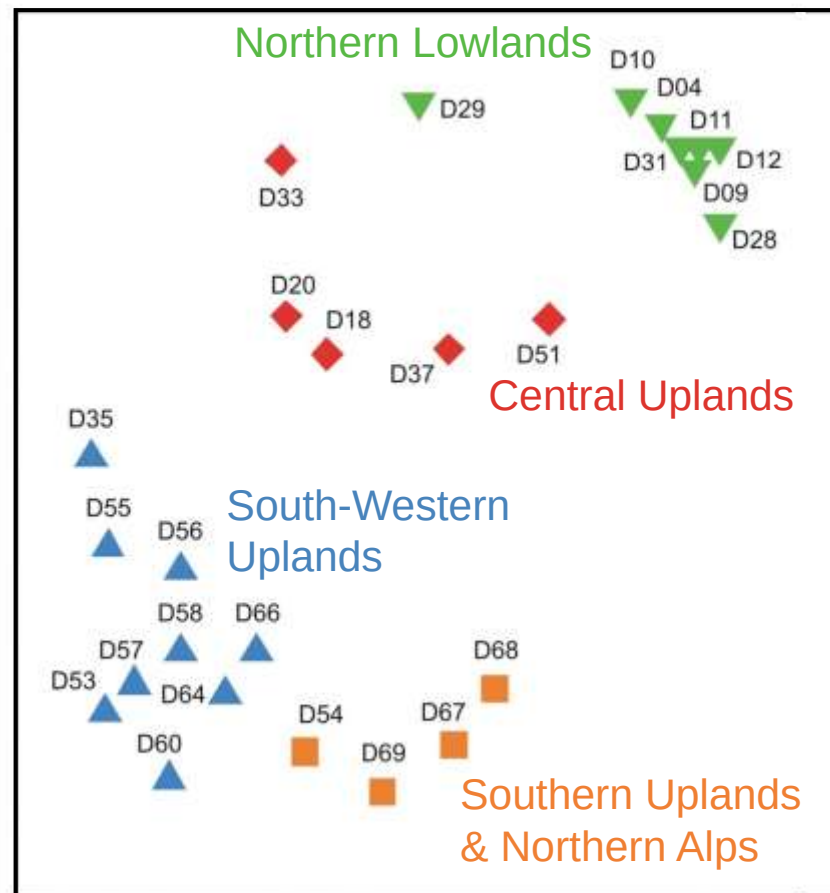
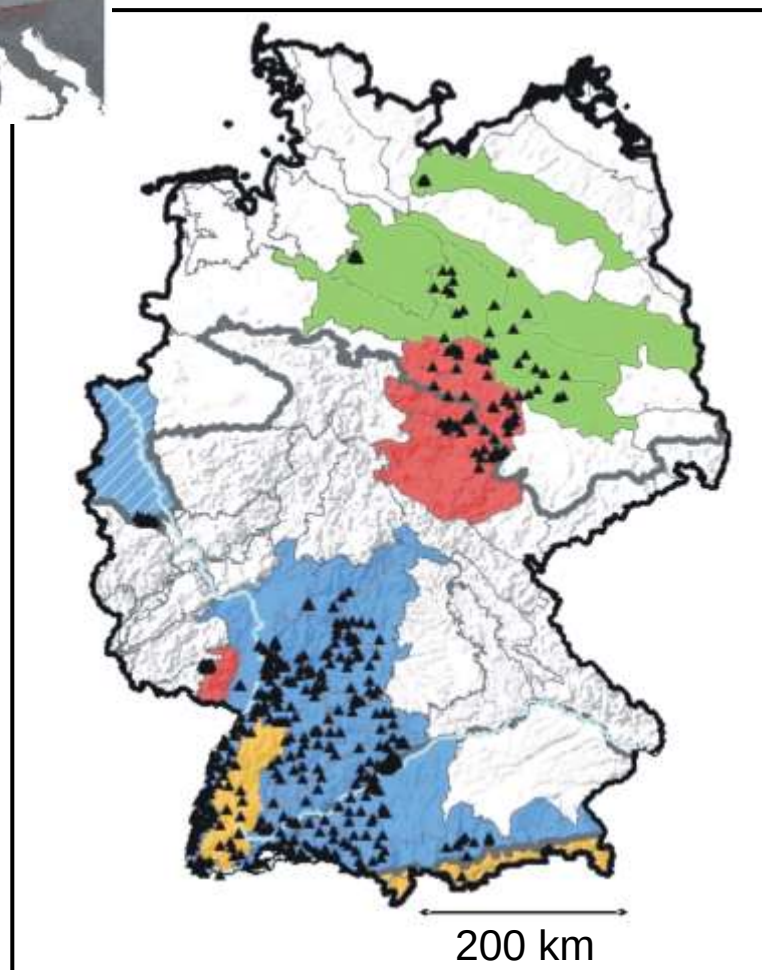


Ferreira et al. (2005)

Community composition: using bioregions (Germany)

26 physiographic units, 514 wells, 1241 samples, 121 species

➔ 4 bioregions



Stein *et al.* (2012)

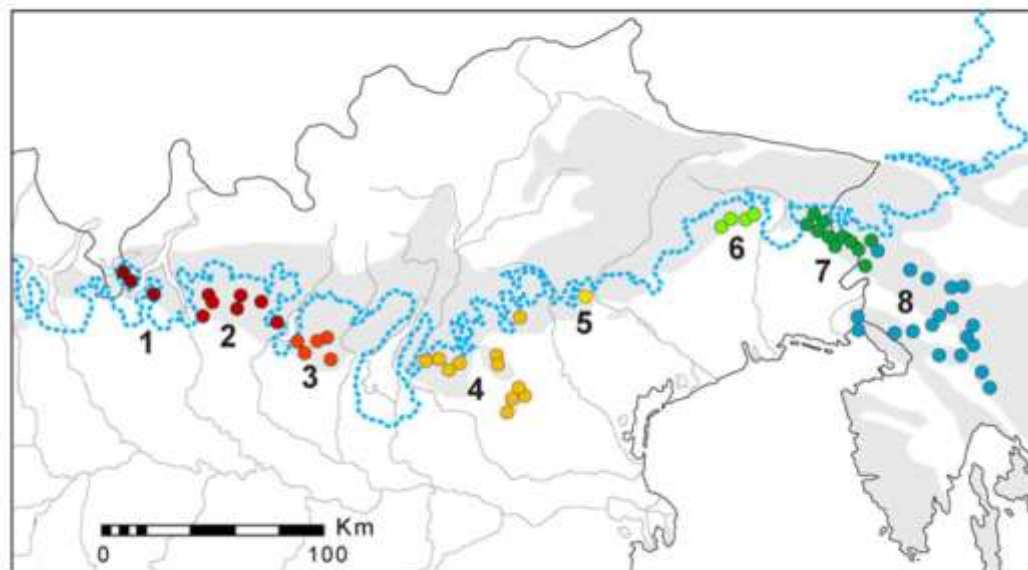
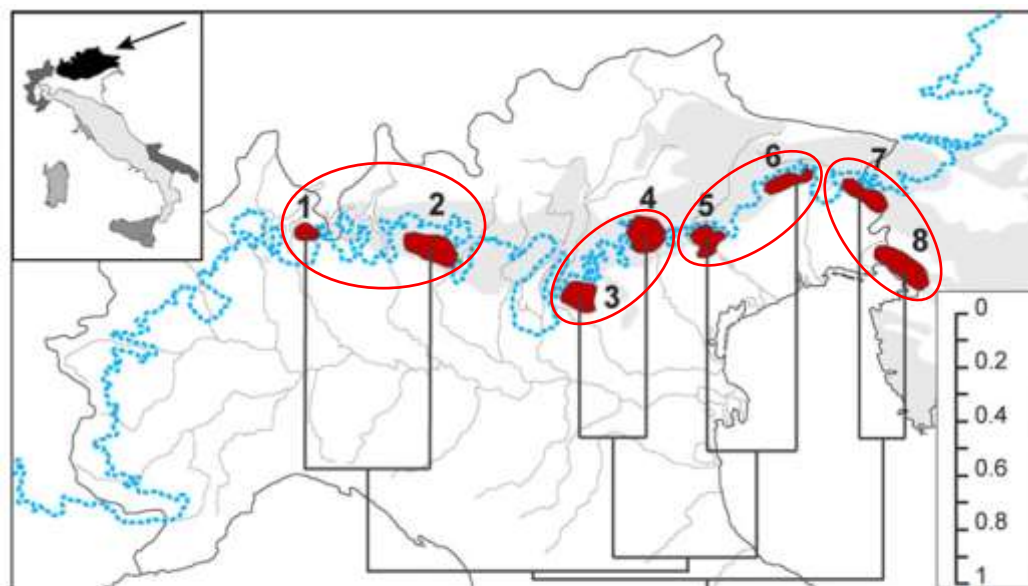


Community composition: using bioregions (Italy)

8 basins, 138 sites,
70 species

➔ 4 bioregions

Set of characteristic
species (*Monolista*)



- | | | | |
|----------------------|---------------------------|---------------------------|-------------------------|
| 1 ● <i>M. pavani</i> | 2 ● <i>M. b. bergomas</i> | 3 ● <i>M. b. boldorii</i> | 4 ● <i>M. berica</i> |
| 5 ● <i>M. sp.</i> | 6 ● <i>M. lavalensis</i> | 7 ● <i>M. julia</i> | 8 ● <i>M. racovitza</i> |

Hydrobiologia
Springer

Stoch & Galassi 2010

Take home messages

- « ~~We know so few~~ »  **PASCALIS** 
- Little scientific constraints for integrating into GWD
 - Habitat typology
 - Habitat-specific biological references
- At EU level with refinements at national level

